MANIPULATING SPINOR INDICES

Theory

Consider the combination $g_{\mu\nu}\sigma^{\mu}_{a\dot{a}}\sigma^{\nu}_{b\dot{b}}$. This carries two dotted and two undotted indices.

These symbols are also invariant. Thus, we can expect the proportionality:

$$\sigma^{\mu}_{a\dot{a}}\sigma_{\mu b\dot{b}} = -2\epsilon_{ab}\epsilon_{\dot{a}\dot{b}} \quad ; \quad \epsilon^{ab}\epsilon^{\dot{a}\dot{b}}\sigma^{\mu}_{a\dot{a}}\sigma^{\nu}_{b\dot{b}} = -2g^{\mu\nu}$$

↳ These are calculated using brute force.

My way of thinking about this is through the LG.

$g_{\mu\nu}\sigma^{\mu}_{a\dot{a}}\sigma^{\nu}_{b\dot{b}} \propto a_{\dot{a}}b_{\dot{b}}$. So when we sum over μ and ν, we are essentially encoding the structure in a way, corresponding to of a $(2,1)\otimes(1,2)\otimes(2,1)\otimes(1,2) = (1,1)_s\oplus$ representation of the Lorentz group.

We are describing the same "physical" object, a spinor set.

We can now try and analyze the generator matrices $(S_L^{\mu\nu})_a{}^b$; $(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{b}}$ in some more detail: $\epsilon_{ab} = L(\Lambda)_a{}^c L(\Lambda)_b{}^d \epsilon_{cd}$

Add this eqn: $L_a{}^b(1+\delta\omega) = \delta_a{}^b + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^b$

That gives us:
$$\epsilon_{ab} = \left\{\delta_a{}^c + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^c\right\}\left\{\delta_b{}^d + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_b{}^d\right\}\epsilon_{cd}$$

$$= \left\{\delta_a{}^b\delta_b{}^d + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^c\delta_b{}^d + \frac{i}{2}\delta_a{}^c\delta\omega_{\mu\nu}(S_L^{\mu\nu})_b{}^d\right\}\epsilon_{cd} + O(\delta\omega^2)$$

$$= \epsilon_{ab} + \frac{i}{2}\delta\omega_{\mu\nu}\left[(S_L^{\mu\nu})_a{}^c\epsilon_{cb} + (S_L^{\mu\nu})_b{}^d\epsilon_{ad}\right] + O(\delta\omega^2)$$

When we contract indices here: $(S_L^{\mu\nu})_a{}^c\epsilon_{cb} = -\epsilon_{bc}(S_L^{\mu\nu})_a{}^c = (S_L^{\mu\nu})_{ab}$ giving us the relation:

$$(S_L^{\mu\nu})_{ab} = (S_L^{\mu\nu})_{ba} \quad ; \quad (S_R^{\mu\nu})_{\dot{a}\dot{b}} = (S_R^{\mu\nu})_{\dot{b}\dot{a}}$$

We can now do a similar analysis for:

$$\sigma^{\rho}_{a\dot{a}} = \Lambda^{\rho}{}_{\tau}L(\Lambda)_a{}^b R(\Lambda)_{\dot{a}}{}^{\dot{b}}\sigma^{\tau}_{b\dot{b}}$$

Here, we have: $\Lambda^{\rho}{}_{\tau} = \delta^{\rho}{}_{\tau} + \frac{i}{2}\delta\omega_{\mu\nu}(S_V^{\mu\nu})^{\rho}{}_{\tau}$; $R_{\dot{a}}{}^{\dot{b}}(1+\delta\omega) = \delta_{\dot{a}}{}^{\dot{b}} + \frac{i}{2}\delta\omega_{\mu\nu}(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{b}}$

$$L_a{}^b(1+\delta\omega) = \delta_a{}^b + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^b \;;$$

Srednicki drops this shit: $(S_V^{\mu\nu}) \equiv \frac{1}{i}(g^{\mu\rho}\delta^{\nu}{}_{\tau} - g^{\nu\rho}\delta^{\mu}{}_{\tau})$

Substituting the above eq^n's yields:

$$(g^{\mu\rho}\delta^{\nu}{}_{\tau} - \underbrace{g^{\nu\rho}\delta^{\mu}{}_{\tau}}_{\downarrow \text{ contract}})\sigma^{\tau}_{a\dot{a}} + i(S_L{}^{\mu\nu})_a{}^b\sigma^{\rho}_{b\dot{a}} + i(S_R{}^{\mu\nu})_{\dot{a}}{}^{\dot{b}}\sigma^{\rho}_{a\dot{a}} = 0$$

multiply by $\sigma_{\rho c\dot{c}}$

we get: $\sigma^{\mu}_{c\dot{c}}\sigma^{\nu}_{a\dot{a}} - \sigma^{\nu}_{c\dot{c}}\sigma^{\mu}_{a\dot{a}} + i(S_L{}^{\mu\nu})_a{}^b\sigma^{\rho}_{b\dot{a}}\sigma_{\rho c\dot{c}} + i(S_R{}^{\mu\nu})_{\dot{a}}{}^{\dot{b}}\sigma^{\rho}_{a\dot{b}}\sigma_{\rho c\dot{c}} = 0$

We can now plug in: $\sigma^{\mu}_{a\dot{a}}\sigma_{\mu b\dot{b}} = -2\epsilon_{ab}\epsilon_{\dot{a}\dot{b}}$

$\rightarrow \sigma^{\mu}_{c\dot{c}}\sigma^{\nu}_{a\dot{a}} - \sigma^{\nu}_{c\dot{c}}\sigma^{\mu}_{a\dot{a}} + 2i(S_L{}^{\mu\nu})_{ac}\epsilon_{\dot{a}\dot{c}} + 2i(S_R{}^{\mu\nu})_{\dot{a}\dot{c}}\epsilon_{ac} = 0$

Now, $\epsilon^{\dot{a}\dot{c}}(S_R{}^{\mu\nu})_{\dot{a}\dot{c}} = -\epsilon^{\dot{c}\dot{a}}(S_R{}^{\mu\nu})_{\dot{a}\dot{c}} = -(S_R{}^{\mu\nu})^{\dot{c}}{}_{\dot{c}}$

We know: $(S_R{}^{\mu\nu})_{\dot{a}\dot{c}} = (S_R{}^{\mu\nu})_{\dot{c}\dot{a}}$

$\rightarrow \epsilon^{\dot{a}\dot{c}}(S_R{}^{\mu\nu})_{\dot{a}\dot{c}} = \epsilon^{\dot{a}\dot{c}}(S_R{}^{\mu\nu})_{\dot{c}\dot{a}} \rightarrow -(S_R{}^{\mu\nu})^{\dot{c}}{}_{\dot{c}} = (S_R{}^{\mu\nu})^{\dot{a}}{}_{\dot{a}} \rightarrow \epsilon^{\dot{a}\dot{c}}(S_R{}^{\mu\nu})_{\dot{a}\dot{c}} = 0$

Multiplying the whole formula by $\epsilon^{\dot{a}\dot{c}}$ once, for a second eq^n by ϵ^{ac}:

$(S_L{}^{\mu\nu})_{ac} = (i/4)\epsilon^{\dot{a}\dot{c}}(\sigma^{\mu}_{a\dot{a}}\sigma^{\nu}_{c\dot{c}} - \sigma^{\nu}_{a\dot{a}}\sigma^{\mu}_{c\dot{c}})$; $(S_R{}^{\mu\nu})_{\dot{a}\dot{c}} = (i/4)\epsilon^{ac}(\sigma^{\mu}_{a\dot{a}}\sigma^{\nu}_{c\dot{c}} - \sigma^{\nu}_{a\dot{a}}\sigma^{\mu}_{c\dot{c}})$

We introduce the new sign: $\bar{\sigma}^{\mu\dot{a}a} \equiv \epsilon^{ab}\epsilon^{\dot{a}\dot{b}}\sigma^{\mu}_{b\dot{b}} = (I, -\vec{\sigma})$

We can then write: $(S_L{}^{\mu\nu})_a{}^b \equiv +(i/4)(\sigma^{\mu}\bar{\sigma}^{\nu} - \sigma^{\nu}\bar{\sigma}^{\mu})_a{}^b$

$$(S_R{}^{\mu\nu})^{\dot{a}}{}_{\dot{b}} = -(i/4)(\bar{\sigma}^{\mu}\sigma^{\nu} - \bar{\sigma}^{\nu}\sigma^{\mu})^{\dot{a}}{}_{\dot{b}}$$

The notation we now take is just to drop spinor indices all together, as in a missing pair of indices is to be assumed. For weyl fields, we have:

$\chi\phi = \chi^a\phi_a$; $\chi^{\dagger}\phi^{\dagger} = \chi^{\dagger}_{\dot{a}}\phi^{\dagger\dot{a}}$

Since weyl fields describe fermions, the fields must anti-commute giving us:

$\chi\phi = \chi^a\phi_a = -\phi_a\chi^a = \phi^a\chi_a = \phi\chi$ Nice and convinient result

Q: How does $-\phi_a\chi^a = \phi^a\chi_a$

We have: $\phi^a = \epsilon^{ab}\phi_b \rightarrow \phi^1 = \phi_2$, $\phi^2 = -\phi_1$

Similarly, $\chi_1 = -\chi^2$, $\chi_2 = \chi^1$

So: $\phi^1\chi_1 + \phi^2\chi_2 = [\phi_2(-\chi^2)] + [(-\phi_1)\chi^1] = -\phi_a\chi^a$

Takeaway: It is worth doing explicit calculations like this to verify and understand things.

We can see that $(2) \longleftrightarrow -(1)$ and this will always be the case for two indices, so ${}^a{}_a = -{}_a{}^a$.

If we take the H.C of $\chi\phi$:

$(\chi\phi)^{\dagger} = (\chi^a\phi_a)^{\dagger} = (\phi_a)^{\dagger}(\chi^a)^{\dagger} = \phi^{\dagger}_{\dot{a}}\chi^{\dagger\dot{a}} = \phi^{\dagger}\chi^{\dagger}$ Very convenient result

Important to keep in mind this works only for two fields.

Taking the H.C of a more complicated term:

$$\phi^\dagger \bar{\sigma}^\mu \chi = \phi^\dagger_{\dot{a}} \bar{\sigma}^{\mu \dot{a} b} \chi_b$$

$$\rightarrow (\phi^\dagger_{\dot{a}} \bar{\sigma}^{\mu \dot{a} b} \chi_b)^\dagger = (\bar{\sigma}^{\mu \dot{a} b} \chi_b)^\dagger (\phi_{\dot{a}}^\dagger)^\dagger$$

$$= \chi_{\dot{b}}^\dagger \bar{\sigma}^{\mu \dot{b} a} \phi_a = \chi^\dagger \underline{\bar{\sigma}} \phi_a$$

Two things to learn here, first, the end result is what we would get on usual †. Second, note that $(\bar{\sigma}^{\mu a \dot{b}})^* = \bar{\sigma}^{\mu \dot{b} a}$ due to hermiticity. I would say, its to preserve the symbol's structure.